UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 4C – Physical Ergonomics: Occupational Biomechanics I

EIN 3241: Human Factors and Ergonomics
Spring 2026

February 13, 2026

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Admin Issues

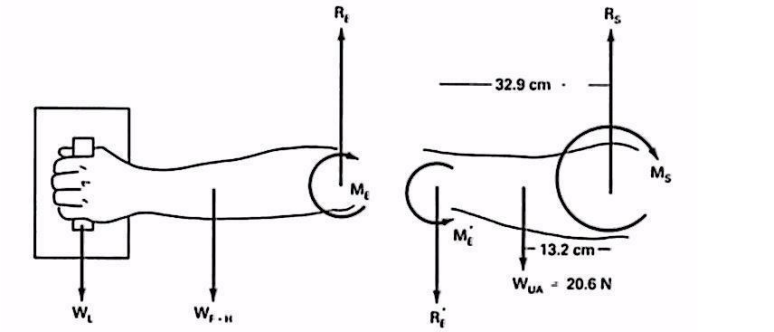
- Lab 2 due Wednesday
- Quiz 3 Wednesday

Learning Objectives

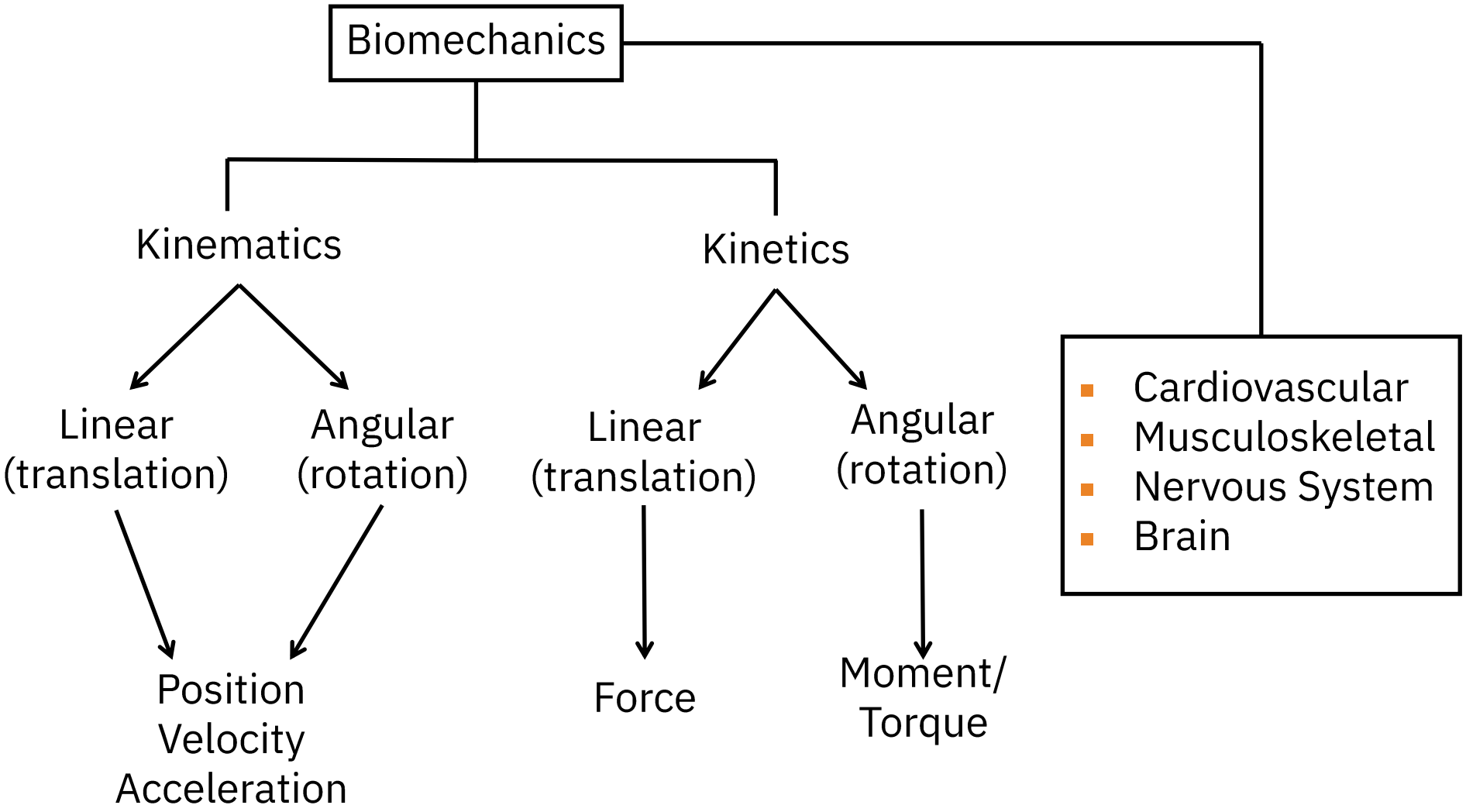
- **Biomechanical approach**: Principles of physics are used to determine the mechanical strain on the body and muscle forces needed to counteract the external stresses
  - Linear and angular kinematics
  - Joint reaction forces
  - Joint reaction moments
- Understanding multibody mechanics for exploring the risk of injury while human interact with products, tools, equipment, and workstation.
- Design of prosthetics, robotics, implants, and other bioengineering products
  - Pre and post movement dynamics

Occupational Biomechanics

- **Biomechanics**: uses the laws of physics and engineering mechanics to describe:
  - The motions of various body segments (**kinematics**) and
  - Understand the effects of forces and moments acting on the body (**kinetics**).
- Two major branches of biomechanics:
  - Kinematics: the study of motion - used in correlation with measurement of body movements (motion capture…). It does not consider "force" component.
  - Kinetics: the study of what causes motion - used in correlation with force / torque measurements (force plates, load cells…). It includes both force and movement.
    - Based measurements which are external to the body.
- **Occupational biomechanics**: studies the physical interaction of workers with their tools, machines, and materials to enhance worker performance while minimizing risk of musculoskeletal injury.



Occupational Biomechanics



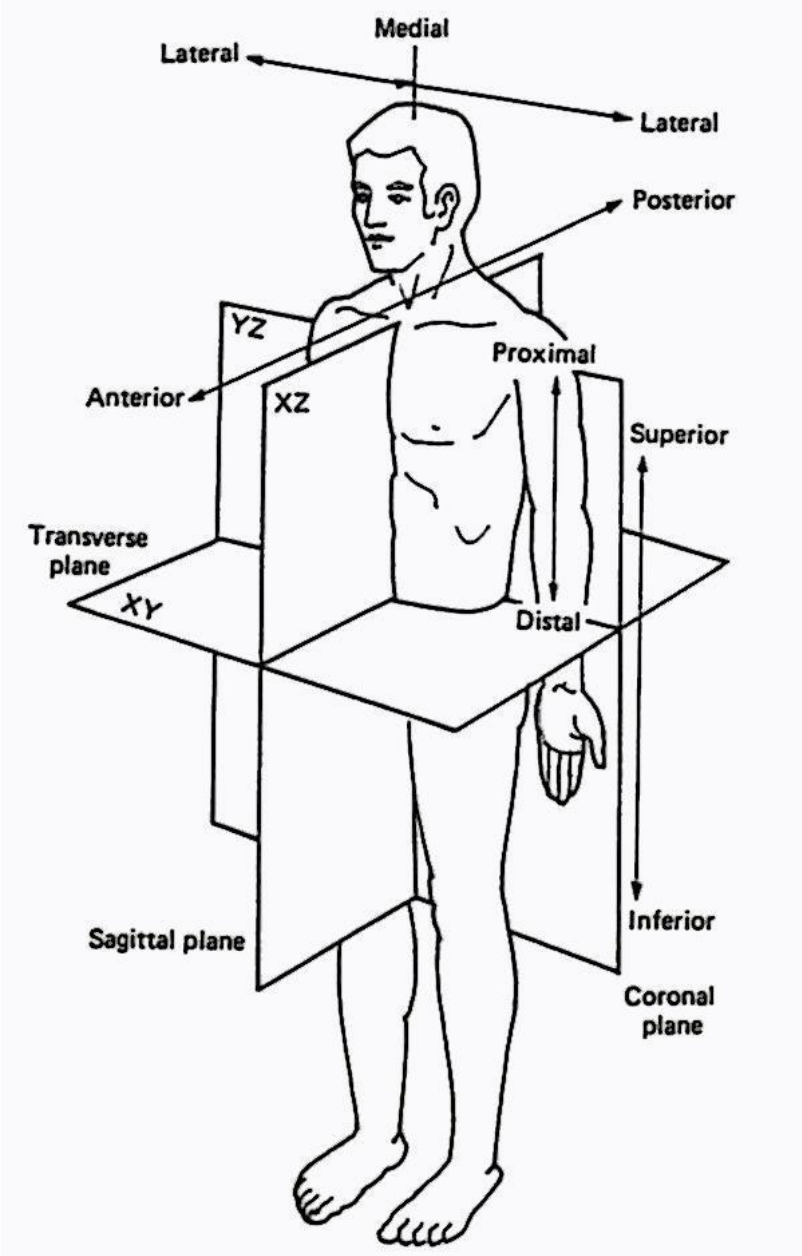
Occupational Biomechanics

- **Biomechanical model**: Smaller scale, simplistic representations of real-life situations for evaluating loads acting on different body segments.

- Potentially dangerous conditions can be evaluated without posing any risk to human health.

Describing body positions/ locations

- Body Planes
  - Sagittal (left and right halves)
  - Frontal (front and back halves)
  - Transverse (upper and lower halves)
- Anatomical Directions/Positions
  - Superior (toward the top)
  - Inferior (toward the bottom)
  - Anterior (toward the front)
  - Posterior (toward the rear)
  - Medial (toward the middle)
  - Lateral (toward the outside)
  - Proximal (closer to the body)
  - Distal (further from the body)
    - (proximal and distal deals with limbs)



Describing body positions / locations (cont'd)

- Standard Anatomical Posture:
  - Standing erect, face forward, holding arms down at side with palms facing forward.

- Other terms used in kinesiology that describe movement:
  - Flexion-decreasing the angle between two bones
  - Extension-increasing the angle between two bones
  - Abduction-moving away laterally from mid-body
  - Adduction-moving toward mid-body
  - Rotation-movement of a bone around its long axis
  - Pronation: face-down, or palm-down position
  - Supination: face-up, or palm-up position

Conventions for Moments and Forces

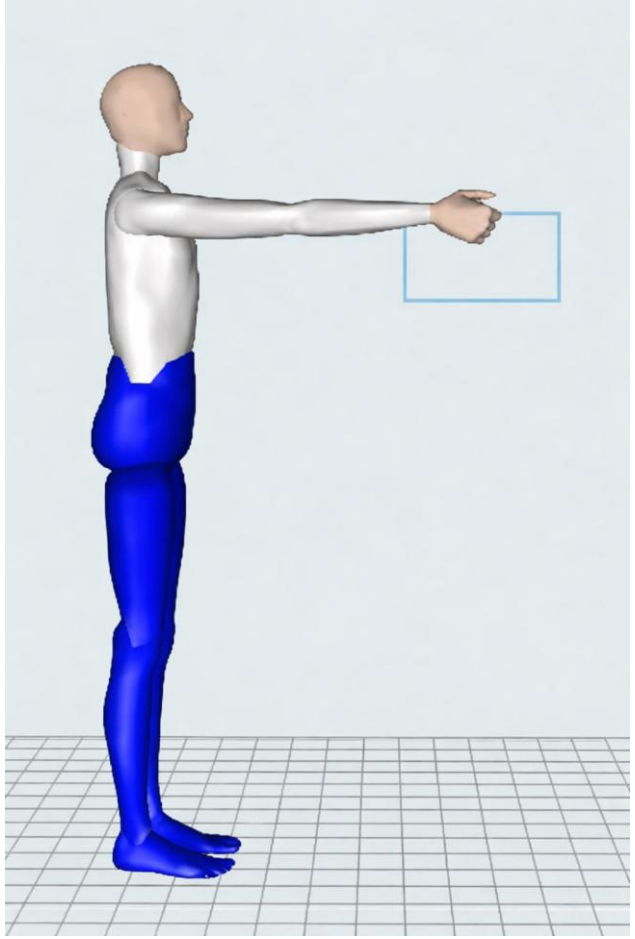
- Equilibrium: If a body is not in motion, the sum of all forces and moments acting on it must be equal to zero.
  - Translational equilibrium: $\sum$ Forces = 0
  - Rotational equilibrium: $\sum$ Moments = 0
- Resultant or External:
  - What the world does to the body (e.g. gravity, contact loading).
- Reactive or Internal:
  - What the body does in response (e.g. muscle activation, ligament stretch).
- In equilibrium: Resultant = Reactive
- Force is measured in Newtons (N), where 1 kg force = 9.81 N

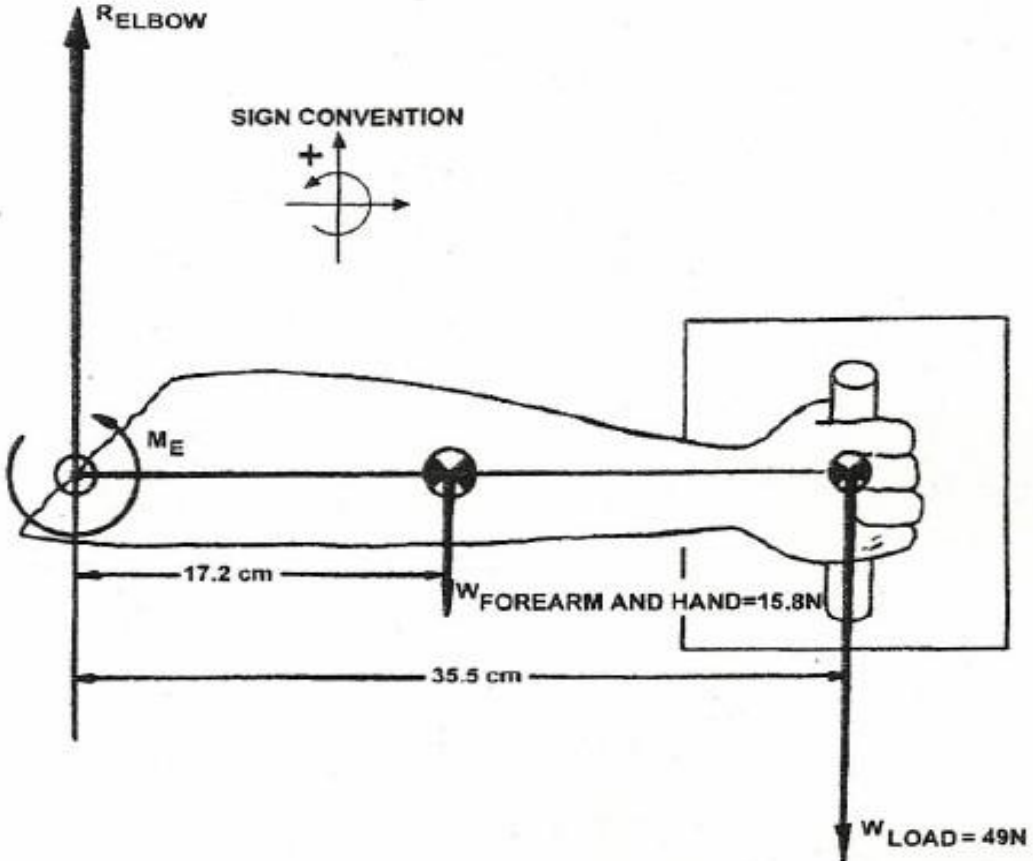
Solving a Biomechanics Model

- There are two approaches to solving a biomechanical model:
  1. Segment-to-segment: treats each body segment as an element in a kinetic chain and works back to all joints affected when handling loads.
  2. One segment: treats all segments as one unit.

Single-Segment Static Model

- Calculate the reaction force and the moment at the elbow joint

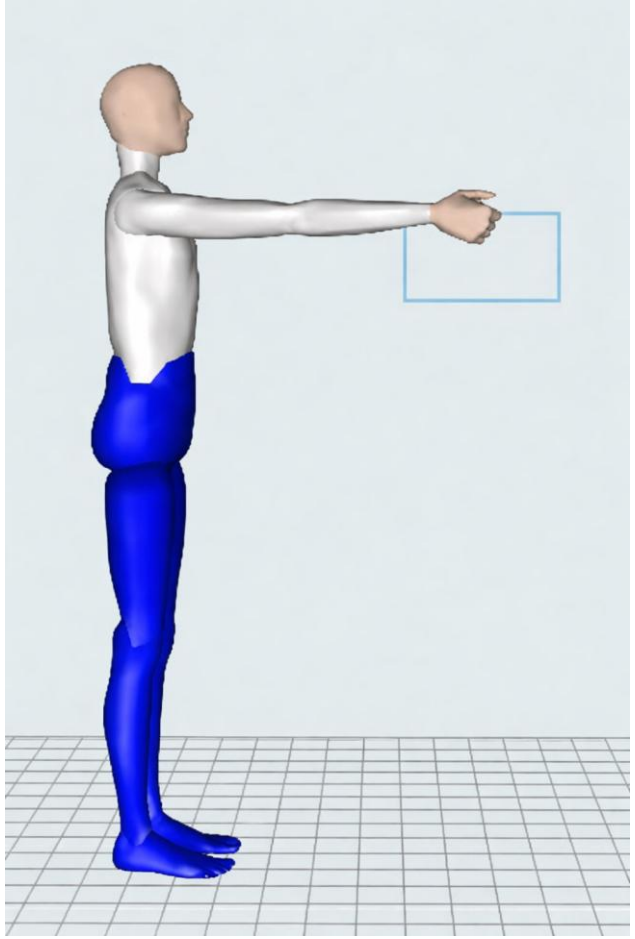


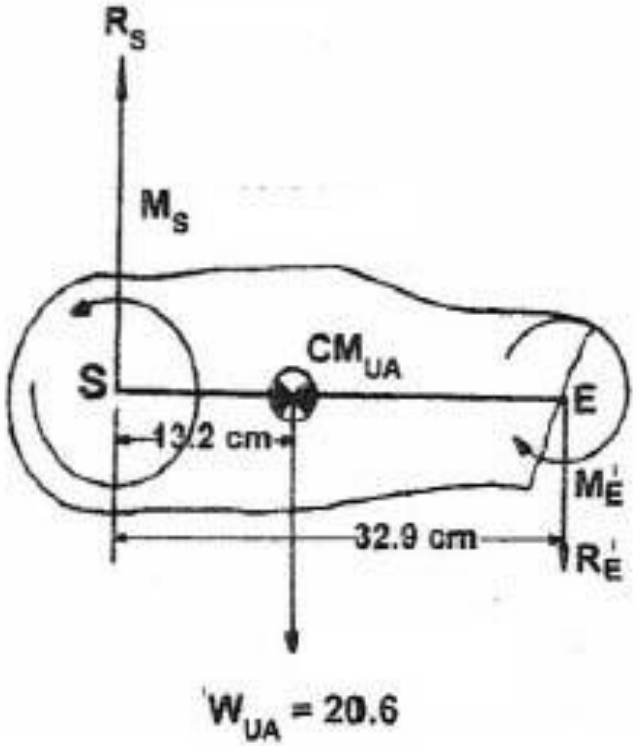


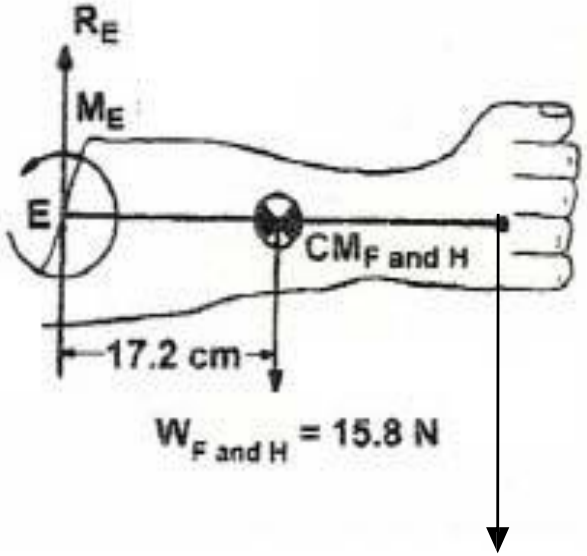
Two Segment Static Model

- Calculate the reaction force and the moment at the shoulder joint

It is possible to treat each body segment as an element in a kinetic chain and work back to all joints affected when handling loads.

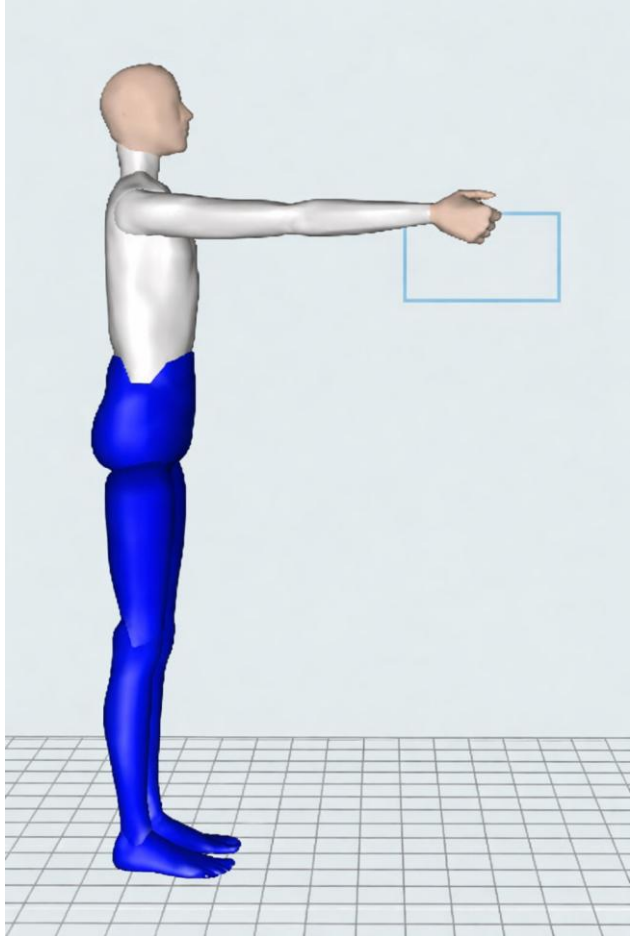






Verify Using One Segment Approach

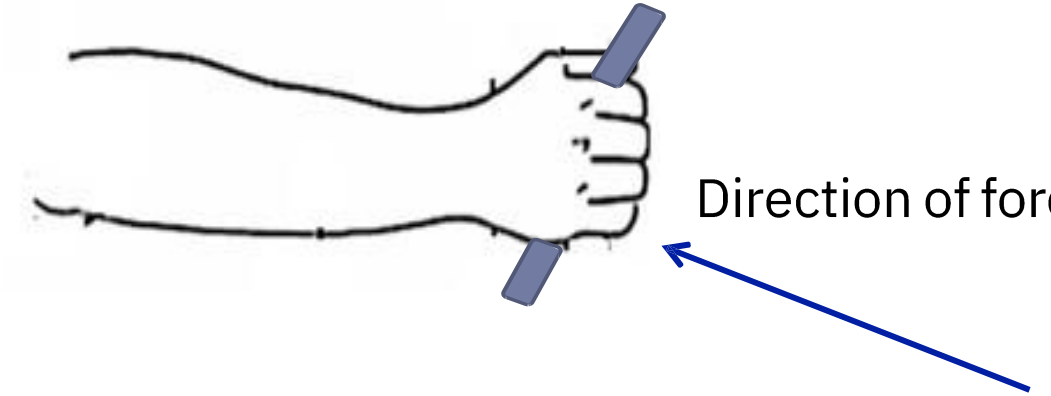
- Calculate the reaction force and the moment at the shoulder joint



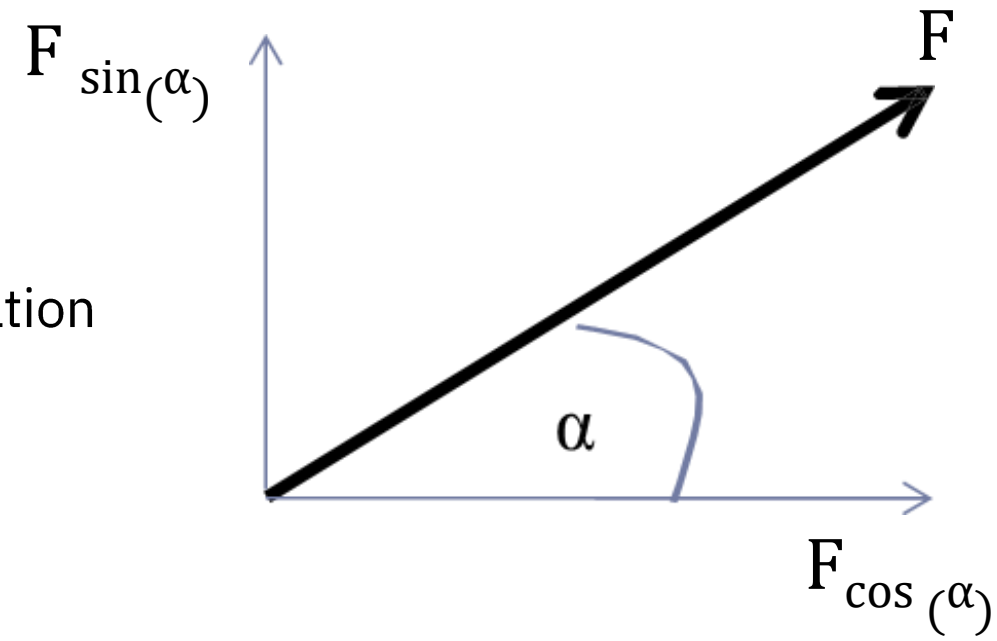
Occupational Biomechanics

- Non-Parallel forces and moment arm
  - Any force can be treated as the vector sum of any two orthogonal forces.

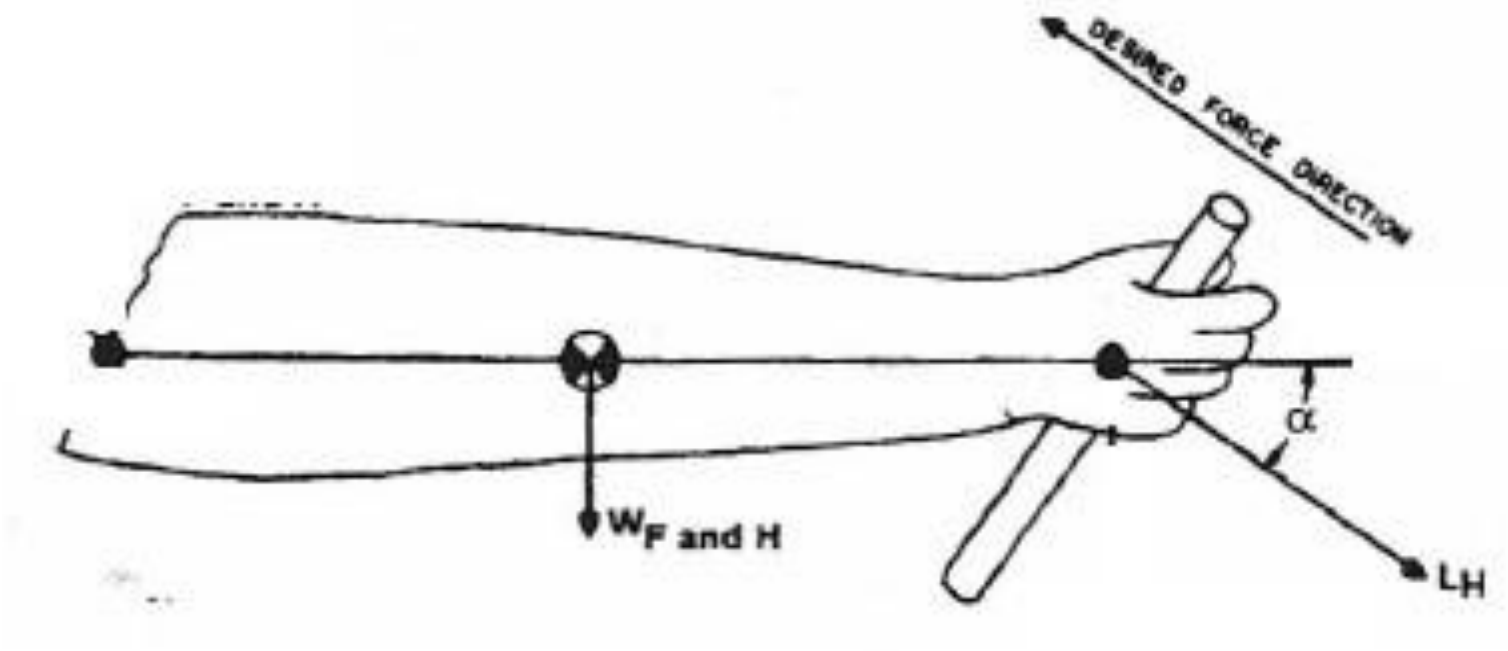
Non-Parallel forces



Direction of force application

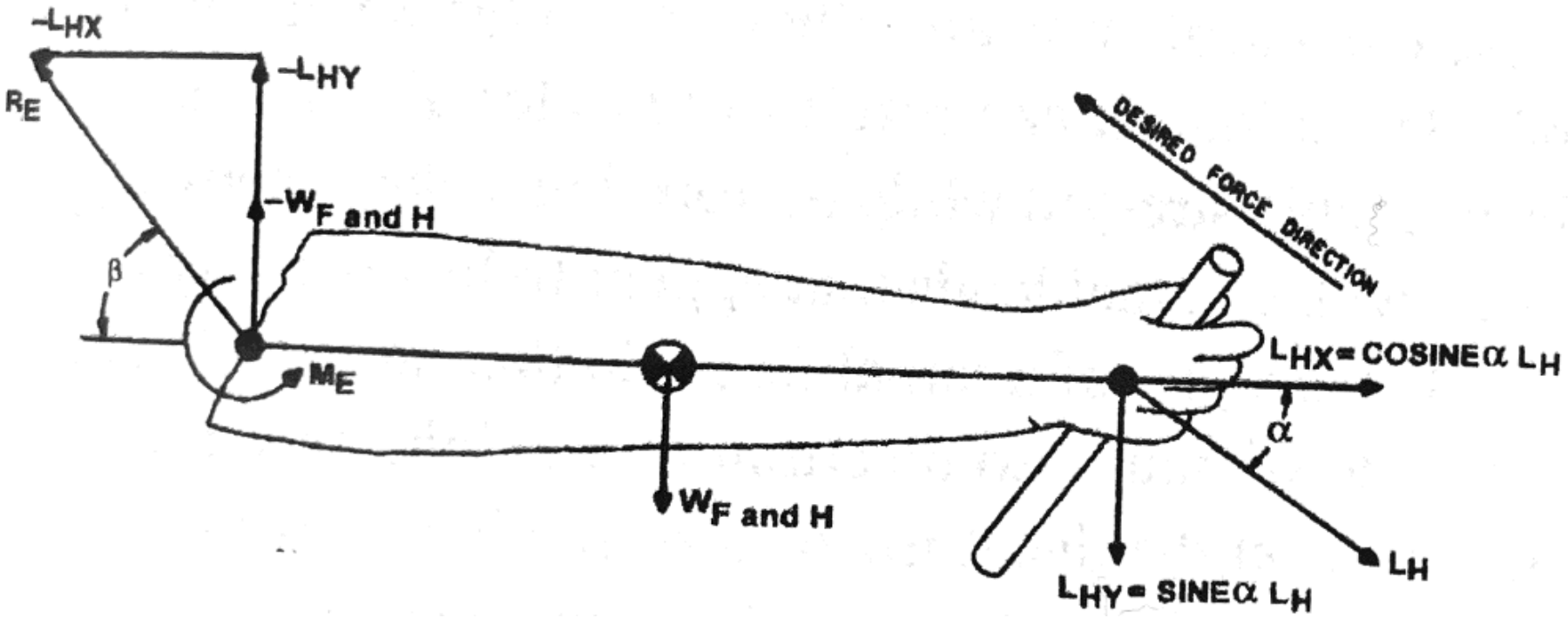


Occupational Biomechanics



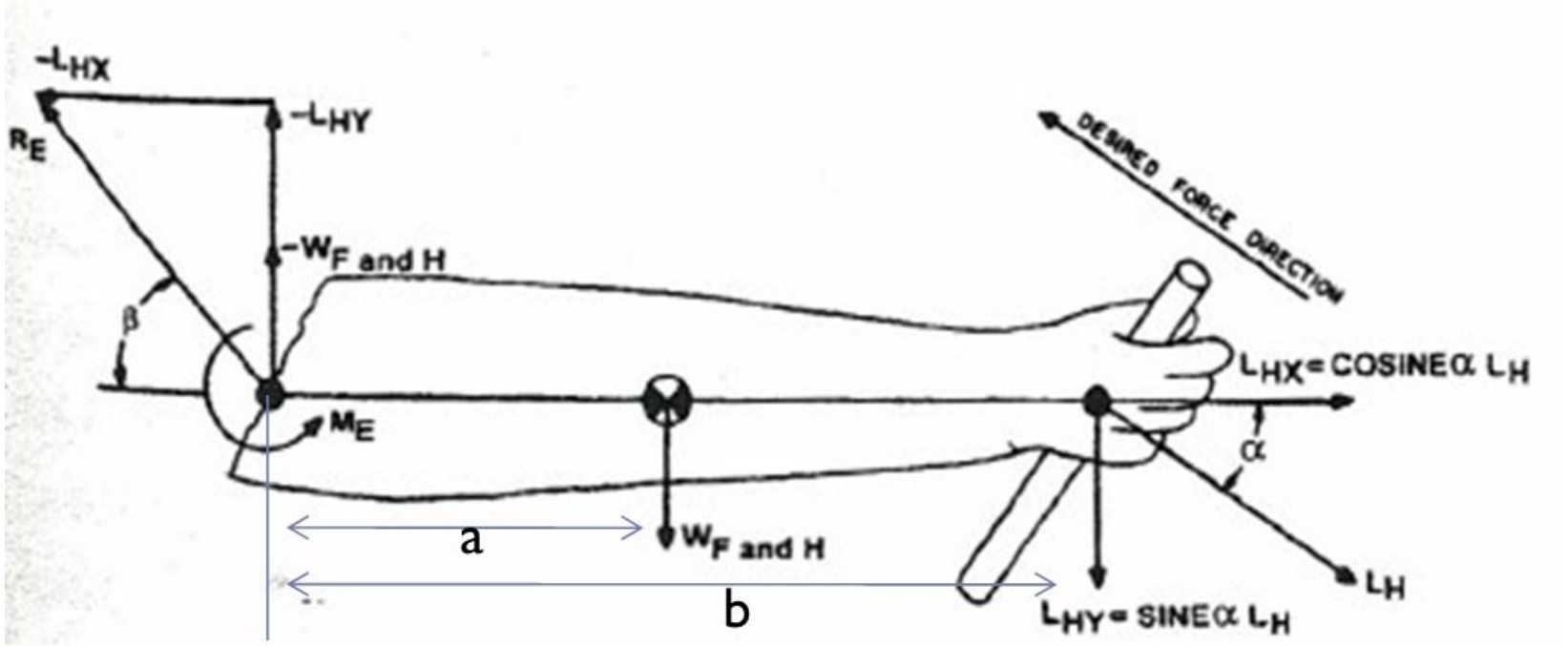
Determine the reaction force and moment at the elbow joint

Occupational Biomechanics



Decomposition of the forces

Occupational Biomechanics



Determine the reaction force and moment at the elbow joint

$$\vec{R}_E = \sqrt{L_{HX}^2 + (L_{HY} + W_{F\&H})^2}$$

$$\beta = \tan^{-1}\left(\frac{L_{HY} + W_{F\&H}}{L_{HX}}\right)$$

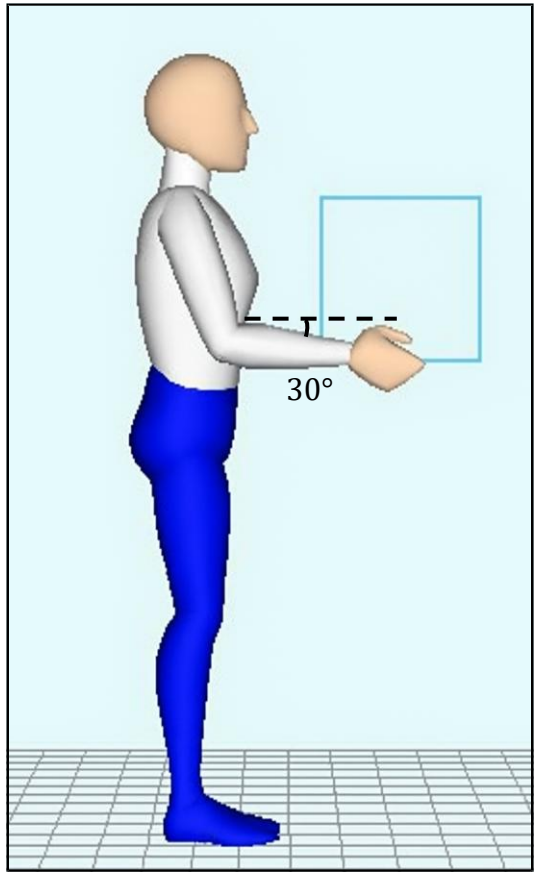
$$\sum M = 0$$

$$\sum M = (L_{HY} \times b) + (W_{F\&H} \times a)$$

What if the elbow is at angle?

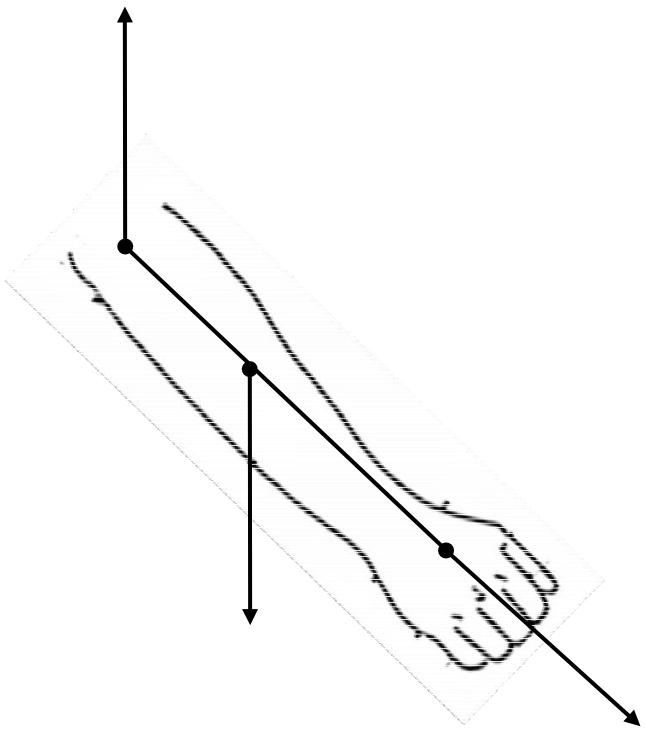
- Calculate the reaction force and the moment at the shoulder joint

It is possible to treat each body segment as an element in a kinetic chain and work back to all joints affected when handling loads.



Another Example

Calculate the reaction force and the moment at the elbow joint for the following pulling task



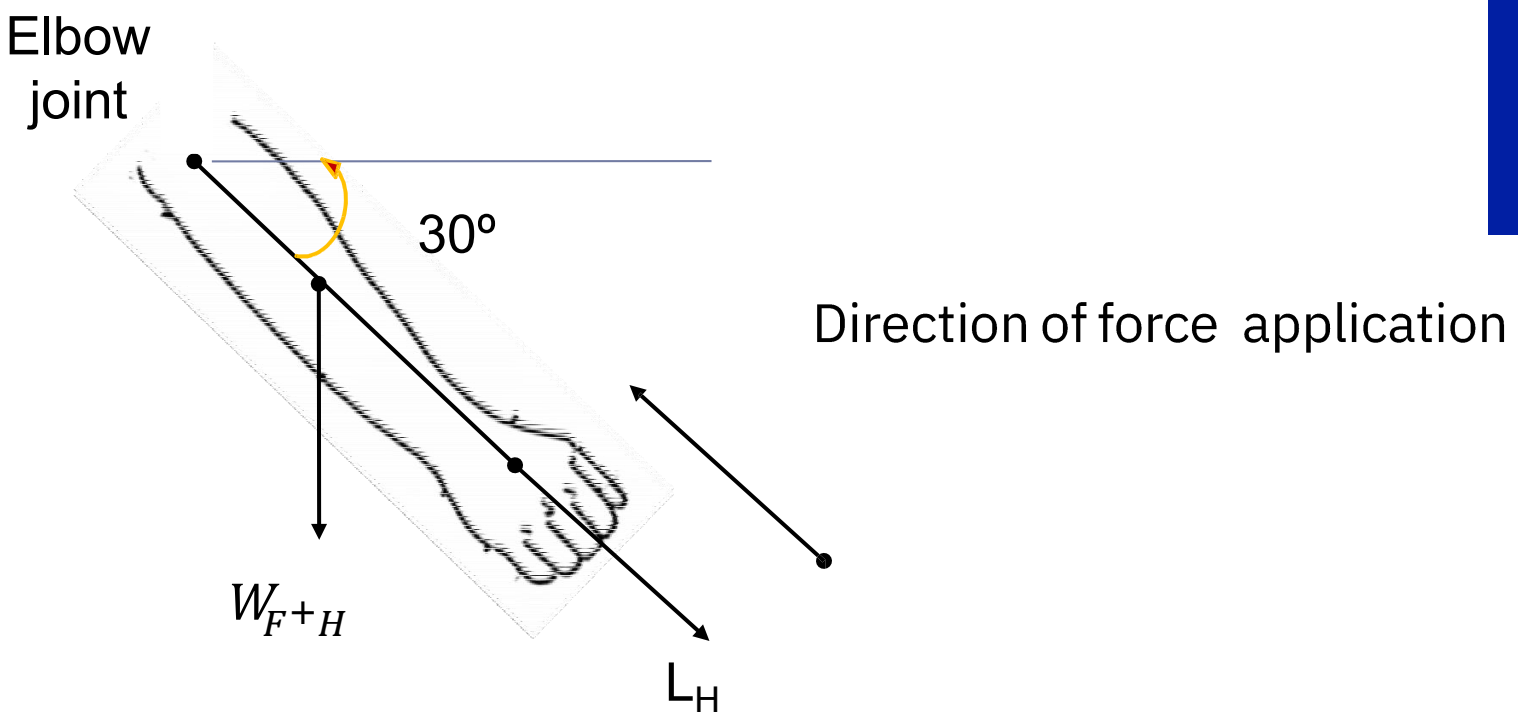
Direction of force application

Weight of forearm and hand = 15.7 N; L_H = 25 kg

Another Example

Calculate the reaction force and the moment at the elbow joint for the following pulling task

Elbow-forearm=0.172m
Elbow-hand=0.355m



Weight of forearm and hand = 15.7 N; L_H = 25 kg

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LEADING THE CHARGE, CHARGING AHEAD

Biomechanics: Useful Tables

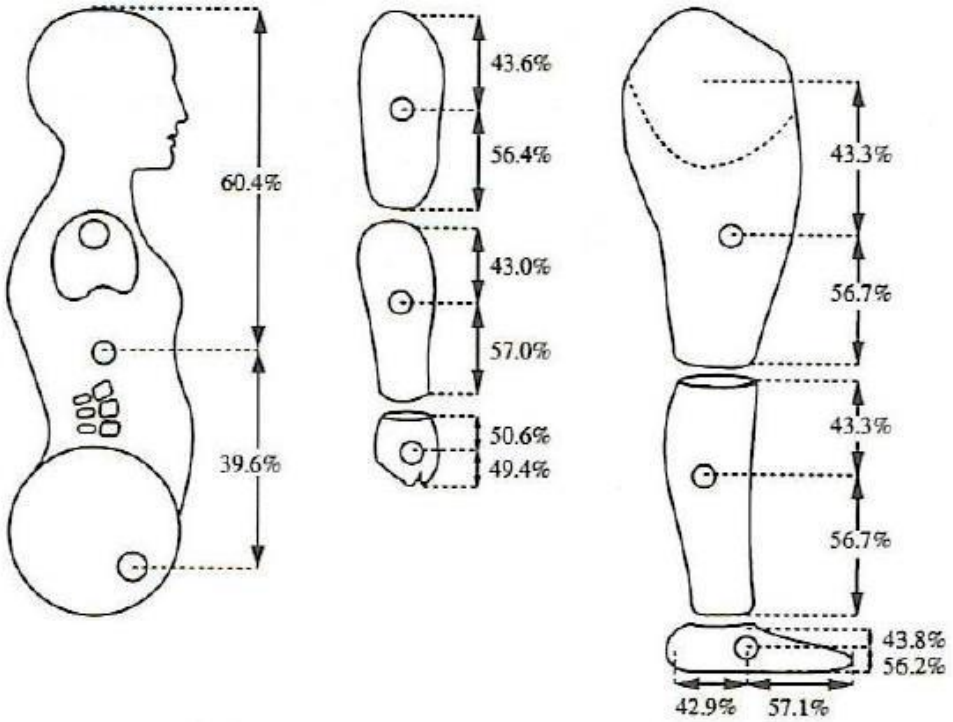


Fig. 4.3 The locations of centers of mass in the body segments in the sagittal plane indicated by percentages of the body segments. Adapted from Dempster (1955).

Table 4.4 Masses of body segments as a percentage of the whole body mass

Group body segments as a percentage of		Individual body segment mass as a percentage of		
	Total body mass (%)		Group segment mass (%)	Total body mass (%)
Head and neck	8.4	Head	73.8	6.2
		Neck	26.2	2.2
Torso (trunk)	50.0	Thorax	43.8	21.9
		Lumbar	29.4	14.7
		Pelvis	26.8	13.4
Each arm (total)	5.1	Upper arm	54.9	2.8
		Forearm	33.3	1.7
		Hand	11.8	0.6
Each leg (total)	15.7	Thigh	63.7	10.0
		Lower leg (shank)	27.4	4.3
		Foot	8.9	1.4

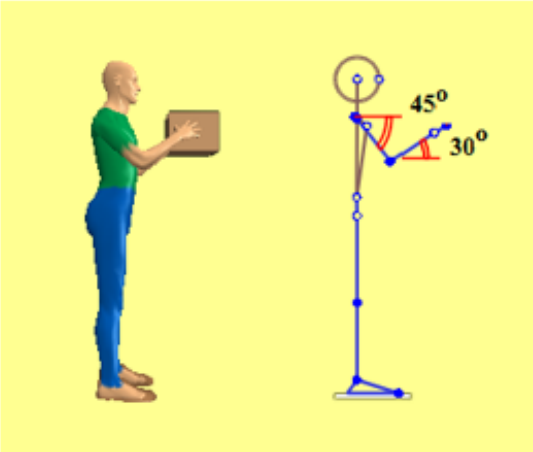
From: Tayyari and Smith (1997). Occupational Ergonomics Principals and Applications. Chapman & Hall: London, pp. 56-57.

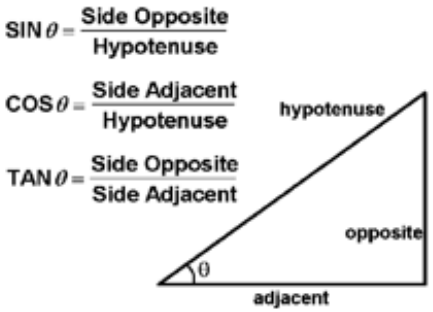
Example

Consider the following information for the lifting task below:

- Mass of box = 15 kg
- Mass of person = 75 kg
- Hand length = 18 cm
- Forearm length = 30 cm
- Upper arm length = 34 cm
- Assume the weight of the box is applied at the center of gravity of the hands.

Determine the reaction moment and forces acting only on the right shoulder due to the upper extremity and weight being lifted.





Solve in two approaches:

1) One Segment Approach
   - Which means the arm as one segment
2) Segment-to-Segment Approach
   - Which means you have to break the arm into segments by joints (wrist, elbow, and shoulder)
   - Along the way, prove that:
     - $R_E = 90.405\ N\ (upwards)$
     - $M_E = 27.78\ Nm\ (CCW)$